

Objective

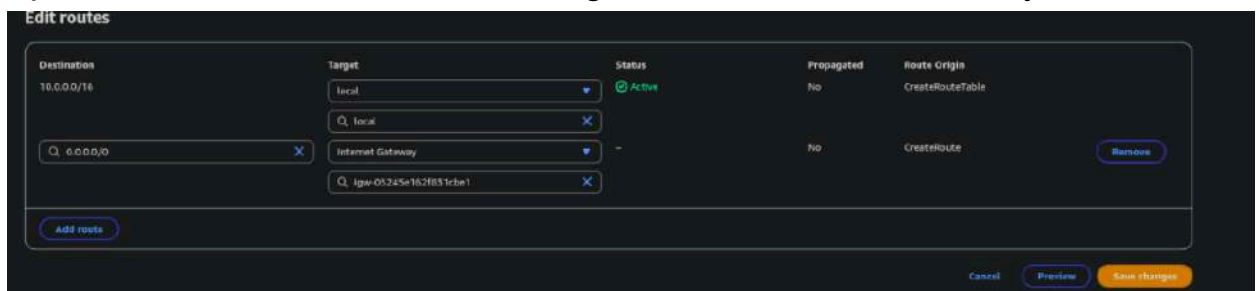
The objective of this lab is to create a custom Amazon VPC with one public subnet and one private subnet, configure networking components such as route tables, an Internet Gateway, and a NAT Gateway, and understand how instances in private subnets can securely access the internet.

Services Used

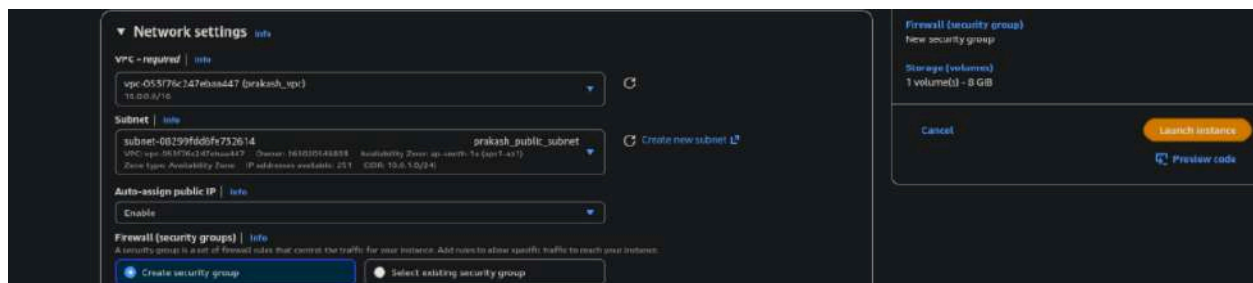
- Amazon VPC
- Amazon EC2
- Internet Gateway
- NAT Gateway
- Route Tables

Create VPC with two subnets one for public and another for private and Internet Gateway

- Go to VPC service
- Create VPC only
- Select subnet then create two subnets
- Create internet Gateway
- Update route table redirect incoming traffic to Internet Gateway



Create 2 EC2 instance one in public subnet and another in private subnet





I take access private instance using public instance :

I first SSH public instance then there i create key pair using from private instance . then i try to ping google.com but did not ping because there isn't attach Internet Gateway.

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-10-0-2-188:~$ ping google.com
PING google.com (142.250.134.193) 56(84) bytes of data.
^C
--- google.com ping statistics ---
119 packets transmitted, 0 received, 100% packet loss, time 120851ms

ubuntu@ip-10-0-2-188:~$
```

i-0dc760c6ae5c4c0c9 (prakash_public_server)

Then i create NAT Gateway for to give access internet to private instance

- Create one more route table for private subnet
- Create Internet Gateway then update route table to incoming any traffic to route NAT gateway
- Confirmed that the NAT Gateway was placed in the public subnet so it could communicate with the Internet Gateway.



When i update route table the i successfully access internet

```
Last login: Wed Jul 1 09:04:22 2020 from 10.0.1.80
ubuntu@ip-10-0-2-100:~$ ping google.com
PING google.com (192.178.174.102) 56(84) bytes of data:
64 bytes from lcboso-in-f102.1e100.net (192.178.174.102): icmp_seq=1 ttl=109 time=2.30 ms
64 bytes from lcboso-in-f102.1e100.net (192.178.174.102): icmp_seq=2 ttl=109 time=1.83 ms
64 bytes from lcboso-in-f102.1e100.net (192.178.174.102): icmp_seq=3 ttl=109 time=1.93 ms
64 bytes from lcboso-in-f102.1e100.net (192.178.174.102): icmp_seq=4 ttl=109 time=1.92 ms
64 bytes from lcboso-in-f102.1e100.net (192.178.174.102): icmp_seq=5 ttl=109 time=1.94 ms
64 bytes from lcboso-in-f102.1e100.net (192.178.174.102): icmp_seq=6 ttl=109 time=1.95 ms
64 bytes from lcboso-in-f102.1e100.net (192.178.174.102): icmp_seq=7 ttl=109 time=1.99 ms
```

I-0dc760c6ae6c4c0c9 (prakash_public_server)
PublicIP: 13.201.58.35 PrivateIP: 10.0.1.80

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Conclusion:

This lab provided practical experience with AWS networking and demonstrated how public and private subnets function within a VPC. It also showed the role of an Internet Gateway in providing internet access to public resources and the importance of a NAT Gateway in allowing private instances to securely access the internet for tasks such as software updates and package installations, without exposing them to inbound internet connections.